The Aperture Firewall: Dual-Wave Closure, Internal Birth-Marks, and Digest-Invisible Structure in the SHA-256 Message Schedule

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Abstract

We report a structural property of the SHA-256 message schedule that has not previously been isolated: the schedule contains a two-rail (dual-wave) topology whose addressability concentrates in a unique pair of positions $K = \{16, 17\}$, and this internal organization is invisible at the digest boundary. Across a sequence of engines (E36–E45) we establish, in this order: (i) $K = \{16, 17\}$ is the universal mixed core of the schedule branch and the bottom of a graded meet-semilattice on 11 residue classes with non-joining even and odd parity horns; (ii) after position normalization, schedule lineage produces no digest-separable class signal ($\eta^2 = 0.000000$ on residualized Hamming distance); (iii) $K = \{16, 17\}$ is the unique minimum sampling aperture for the dual wave — among all pair-removals of W[i], W[j] with $16 \le i < j \le 21$, only the K-pair produces the balanced collapse signature ($D = 0$, both horns dead, 11 residue classes preserved); (iv) the firewall is scaffold-stable, not a carry artifact: K is invisible in both real SHA-256 (Track A) and the carry-free GF(2) shadow (Track B), with $|Z| < 0.6$ in each. The strongest result is the residual one: the carry-exhaust differential $\Delta = A - B$ detects K at $Z(t = 16) = -3.33$. The carry channel records the seam even though the digest boundary erases it. We name this phenomenon the aperture firewall: a system boundary at which internal organization remains indispensable to correct operation while external readout remains intentionally uniform. The result is not a weakness in SHA-256; it is an explicit characterization of how the standard's diffusion machinery achieves what it claims. We close by

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extending the aperture-firewall concept to AI governance, where systems evolving faster than human inspection require trust to migrate from direct observation to disciplined assurance: monitored invariants, bounded intervention, and verified internal apertures.

1. Introduction

NIST FIPS 180-4 specifies SHA-256 as a two-stage process of preprocessing and hash computation. Hash computation generates a 64-word message schedule $W_0, \ldots, W_{63}$, updates eight working variables a, ..., h across 64 rounds, and outputs a 256-bit digest. The standard explicitly states that a change to the message should, with very high probability, produce a different message digest. That design goal is a diffusion objective, and the empirical literature has long treated avalanche-style behavior as a manifestation of confusion-and-diffusion quality.

This paper asks a narrower and more structural question. If the schedule contains a forced internal organization, does that organization leak into the digest as a class-separable signal, or does the compression boundary erase it? We answer the second outcome is the correct one, and we make the structure of the question precise. The schedule contains a graded meet-semilattice over 11 residue classes; this lattice has a uniquely defined bottom — the seam K = {16, 17} — and two non-joining parity horns. The lattice is internally load-bearing in the strict sense that damaging K collapses the dual-rail topology while damaging arbitrary downstream positions does not. Yet the digest avalanche on K-damage is statistically indistinguishable from the digest avalanche on arbitrary-position damage.

The paper formalizes this as the aperture firewall theorem and extends it from cryptographic compression to AI governance. The bridge is direct: machine-speed systems evolve faster than human operators can directly inspect every internal state transition. NIST's AI RMF defines trustworthy AI in terms of validity, reliability, safety, resilience, accountability, transparency, and ongoing risk management rather than perfect observability. Aperture trust is what we call the principled stance that internal load-bearing structure can be governed by monitored invariants and bounded escalation, rather than by the impossible demand of microstate visibility.

2. Background

2.1 SHA-256 Schedule

For a padded 512-bit block parsed into sixteen 32-bit words, the message schedule extends by

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}}$$

$$\text{for } 16 \le t \le 63$$

and the round update uses

$$T_1 = h + \Sigma_1(e) + Ch(e,f,g) + K[t] + W[t]$$

$$T_2 = \Sigma_0(a) + Maj(a,b,c)$$

$$a' = T_1 + T_2, \quad e' = d + T_1$$

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followed by the standard state rotation. These are the operations that create the tension at issue: the schedule has its own combinatorial geometry, but the compression rounds are built to diffuse and scramble perturbations rapidly.

2.2 Lineage Topology

For each W[t] we define its lineage as the set of seed indices {0, ..., 15} on which W[t] transitively depends through the recurrence. For seeds $t < 16$, lineage(W[t]) = {t}. For $t \geq 16$, lineage(W[t]) is the union of lineages of its four predecessors. The collection of distinct lineages over $t = 16, \ldots, 63$, ordered by set inclusion, forms a meet-semilattice. Two lineages have a meet (greatest lower bound, set intersection), but not every pair has a join (least upper bound) realized as a W[t] lineage.

Direct computation gives lineage(W[16]) = {0, 1, 9, 14} and lineage(W[17]) = {1, 2, 10, 15}. These are the first mixed-seed lineages and constitute the bottom of the mixed-region lattice. The lineages of W[18] = {0, 1, 2, 3, 9, 11, 14} and W[19] = {1, 2, 3, 4, 10, 12, 15} are the first atoms above the bottom — minimal lineages strictly containing one of the bottoms. We label these atoms R8 = lineage(W[18]) and R9 = lineage(W[19]).

2.3 Dual-Rail Score

To measure whether the dual-rail topology survives an intervention we define a three-component score:

$$D = \mathbb{1}[R8 \text{ exists}] + \mathbb{1}[R9 \text{ exists}] + \mathbb{1}[R8 \vee R9 \text{ does not exist in the lattice}]$$

D = 3 indicates: two atoms exist; the atoms separately cover the seam; and there is no W[t] whose lineage equals lineage(R8) ∪ lineage(R9). This last condition is what makes the horns non-joining — the rails cross only by acquiring additional seed material, so the pure union is never realized as a single word's lineage. D = 0 indicates that the dual rail has collapsed either by losing both atoms, by losing the non-joining condition, or by losing the rail-separating structure.

3. Prior Results (E36–E43)

Five prior engines established the structural object on which this paper builds. We summarize them with the minimum necessary detail; full reports are in the QuHarmonics A-Mark9 phase logs.

3.1 E36 — K = {16, 17} as Universal Mixed Core

Across all mixed seed-pair intersections in the schedule branch, 36 of 36 contained the index pair {16, 17}, and factoring out the seam left an 11-class residue structure. The same engine showed that parity-crossing always occurred after lag 7. The seam is not special because it crosses faster; it is special because it is positionally first.

3.2 E37 — GL(4, ℂ) Rejected; Wheel-210 Recurrence Confirmed

An attempted GL(4, ℂ) interpretation of the seam alphabet did not survive computation. The living structure is a GF(2)-linear recurrence tied to wheel-210 geometry: radical 210, $\omega(210) = 4$, $\phi(210) = 48$ — the same size as the SHA-256 expansion zone. The paper holds the GL(4, ℂ) gate closed on this finding.

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3.3 E38 — Meet-Semilattice with Non-Joining Horns

The 11-class residue alphabet is not a lattice and not a group orbit. It is a graded meet-semilattice with a bottom seam, two parity atoms above the bottom, and non-joining maximal even and odd horns. The non-joining property is precisely the lattice statement that the join R8 ∨ R9 is not realized.

3.4 E39 and E40 — Horns Operator-Separated; Birth-Mark Persists Internally

Cross-horn σ_1 edges are absent at the relevant scales, and phased entanglement begins at W[23]. The schedule retains a stable parity birth-mark across all words, but only as an internal property of the lineage object.

3.5 E41 — Birth-Mark Internal-Only at Compression Boundary

Carrying the schedule's parity ancestry forward into the compression machinery and reading the digest, the birth-mark does not appear as a class-separable signal. Real SHA and a partial GF(2) shadow both produced strong diffusion. The seam is internally real but digest-invisible.

3.6 E42 — Corpus Stress Test

Across a broader corpus the earlier W[47] anomaly did not replicate. A modest raw class effect remained, $\eta^2 \approx 0.028$, but engine output identified schedule position as the real driver: late schedule words have fewer compression rounds remaining before feed-forward, so their average digest Hamming distances are lower regardless of class.

3.7 E43 — Position-Normalized Class Effect Vanishes

Define the position-normalized residual

$$r(m, t, b) = HD(m, t, b) - E[HD \mid t]$$

where HD(m, t, b) is the digest Hamming distance from flipping bit b of W[t] in message m. After this normalization the class effect on residuals is exactly zero: $\eta^2 = 0.000000$ with maximum |mean residual| = 0.0000 bits across all lineage classes. The empirical theorem is

$$lineage(W[t]) \Longrightarrow no\ digest\ class\ signal \mid t$$

with all raw class effects identified as position confounds.

4. Engine 44 — Dual-Wave Minimum Aperture

E43 established the digest-class firewall. The natural next question is whether the schedule's internal dual-rail structure can itself be characterized as a minimum aperture for the wave. We show it can, and that K = {16, 17} is the unique pair satisfying the characterization.

4.1 Protocol

Three intervention classes are applied to the lineage recurrence:

1. Topological removal: lineage(W[t]) := ∅
2. Identification: lineage(W[j]) := lineage(W[i])
3. Value removal (test): W[t] := 0 in real SHA execution

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Controls: seed perturbations (W[5], W[8]), downstream removals (W[25], W[30], W[40], W[50], W[60]), and downstream identifications (W[25] := W[23], W[50] := W[48]). For each intervention we compute the dual-rail score D and the structural fingerprint (number of atoms, number of residue classes, even-horn size, odd-horn size, R8 ∨ R9 no-join status).

4.2 Result 1: K-damage uniquely collapses the dual rail

Direct K-targeted interventions:

Intervention	D	atoms	classes	h_even	h_odd	no-join
BASELINE (K intact)	**3**	2	11	3	2	YES
Remove W[16]	1	2	11	0	43	no
Remove W[17]	0	1	11	44	0	no
Remove both W[16], W[17]	**0**	2	11	0	0	no
Identify W[17] := W[16]	**0**	2	11	0	0	no

Every K-targeted intervention drops D from 3 to ≤ 1. The full collapse signature (D = 0, both horns dead, 11 residue classes preserved) arises specifically in the symmetric K-damage cases — remove both, or identify the two. The triangle reading is direct: W[16] and W[17] are the two BUILT vertices of the dual wave, and the non-joining condition R8 ∨ R9 ∉ lattice is the GAP that constitutes the third vertex of closure. Identifying W[17] := W[16] eliminates the gap directly; the triangle collapses to a line; D = 0.

4.3 Result 2: Downstream perturbations preserve the dual rail

Of the 64 schedule positions, removal of any of 53 leaves D = 3 unchanged. The critical set is exactly 11 positions:

Critical: $t \in \{0, 2, 9, 10, 14, 15, 16, 17, 18, 19, 20\}$ (11 positions)

Safe: $t \in \{1, 3, 4, 5, 6, 7, 8, 11, 12, 13, 21..63\}$ (53 positions)

The critical set partitions naturally into seam feeders, the seam K itself, the rail atoms R8 and R9, and the immediate downstream extension W[20]:

Subset	Members	Role
Seam feeders	{0, 2, 9, 10, 14, 15}	Asymmetric seed inputs to K
K (seam)	**{16, 17}**	**Minimum dual-wave aperture**
First atoms	{18, 19}	R8, R9 — rail roots
Atom downstream	{20}	Even-rail extension

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The critical seed set excludes only W[1], which is the unique seed common to both L_{16} and L_{17}. Removing W[1] preserves D = 3 because the rails still differ. Removing any seed that lies in exactly one of L_{16} or L_{17} damages the rail asymmetry.

4.4 Result 3: K = {16, 17} is the unique minimum aperture

We sweep all pair-removals (i, j) with 16 ≤ i < j ≤ 21 and tabulate the structural fingerprint (D, atoms, classes, h_even, h_odd):

pair	D	atoms	classes	h_e	h_o	signature
(16,17)	**0**	2	11	0	0	**(0,2,0,0,11) ★ unique**
(16,18)	0	1	10	0	43	(0,1,0,43,10)
(17,18)	0	2	12	38	0	(0,2,38,0,12)
(18,19)	1	2	13	4	3	(1,2,4,3,13)
(19,20)	3	2	13	4	3	(3,2,4,3,13) — dual rail intact

Among 15 pair-removals tested in this region, K = (16, 17) is the only pair producing the signature (D = 0, atoms = 2, h_even = 0, h_odd = 0, classes = 11). Every other pair either kills only one horn (asymmetric collapse) or leaves D ≥ 1 with at least one rail surviving. The balanced collapse — both horns simultaneously dead while the residue alphabet is preserved at the canonical 11 classes — is uniquely the K-signature.

4.5 Result 4: Numerical avalanche is uniform across interventions

In parallel we ran value-removal (W[t] := 0) on the same interventions in real SHA-256 across a 200-message random corpus. Per-position mean digest Hamming distance:

t = 0 → HD = 128.28 (50.1%)
t = 16 → HD = 127.09 (49.6%) ◀ K
t = 17 → HD = 127.38 (49.8%) ◀ K
t = 18 → HD = 129.27 (50.5%)
t = 40 → HD = 128.59 (50.2%)
t = 63 → HD = 32.05 (12.5%) ← diffusion-horizon collapse

Fitting a degree-3 position baseline E[HD | t] and computing |HD − E[HD | t]| residuals:

|residual| at K = (16, 17): mean = 3.20
|residual| elsewhere: mean = 6.26
|residual| elsewhere: max = 56.37 (t = 63)

K = {16, 17} sits below the elsewhere mean in residual magnitude — not above. The structural collapse documented in Results 1–4 is invisible at the digest boundary. This is the aperture firewall at fold scale.

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5. Engine 45 — Carry-Shadow Firewall Test

E44 left one explicit open line: whether the digest uniformity under K-damage originates in the linear schedule scaffold or in the carry arithmetic of add_{32}. E45 closes that line by running every E44 intervention in two parallel tracks.

5.1 Track A versus Track B

Track A is real SHA-256 with add_{32} carries intact. Track B is the GF(2) shadow: every modular addition in the schedule recurrence and the round function is replaced by XOR, while the bitwise nonlinear gates Ch and Maj are retained. Both tracks share the same K constants, initial vector, and circuit topology. The only difference is whether arithmetic carries propagate.

If K is invisible in Track B as well as Track A, the firewall is scaffold-level — the linear skeleton already produces uniformity at K. If K is visible in Track B but hidden in Track A, the firewall is carry-stabilized — the linear skeleton leaks, and the carry channel actively erases the leak.

5.2 K-damage in both tracks (single block)

With a fixed reference block, we record digest Hamming distance against baseline in both tracks:

Intervention	HD_A	HD_B	Δ (A−B)
BASELINE	0	0	0
Zero W[16]	124	128	−4
Zero W[17]	119	120	−1
Zero both W[16], W[17]	130	134	−4
Identify W[17] := W[16]	109	129	**−20**
CTRL Zero W[25]	126	125	+1
CTRL Zero W[40]	118	117	+1
CTRL Zero W[50]	138	124	+14

At the single-block scale the K-damage Δ values are negative — Track A produces slightly less avalanche than Track B at K. Controls span both signs. To separate signal from message-specific variation we run the position sweep.

5.3 Per-position residualization

Over N = 200 random messages, for each $t \in \{0, \ldots, 63\}$ we zero-inject W[t] in both tracks and compute mean HD. We then fit a degree-3 position baseline E[HD_A | t] and E[HD_B | t] and examine residuals at K relative to the elsewhere distribution.

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Channel	|K| mean	|elsewhere| mean	Z(t=16)	K outlier?
Track A (real SHA-256)	3.22	6.26	+0.36	No
Track B (GF(2) shadow)	4.19	6.31	+0.51	No
Δ = A − B (carry exhaust)	0.96	0.51	**−3.33**	**Yes**

5.4 Interpretation

The result is sharper than the binary scaffold-level / carry-stabilized disjunction the protocol named. K is not a numerical outlier in Track A (|Z| = 0.36) and is not a numerical outlier in Track B (|Z| = 0.51). The firewall is present in both tracks individually. But the difference channel Δ = A − B carries a K-specific signature with Z(t = 16) = −3.33, exceeding the conventional outlier threshold at over three standard deviations.

The mechanism is double-layered. Both the GF(2) scaffold and the carry channel independently produce uniform avalanche under K-damage. The K position is the locus where their internal contributions differ measurably: Track B's linear skeleton would produce a slightly elevated residual at K, and the carry channel of Track A subtracts that excess, leaving the two tracks at indistinguishable totals. The carry exhaust thus 'knows about K' even though the digest does not.

The full firewall theorem now reads:

The aperture firewall is scaffold-stable and carry-reinforced. The linear skeleton alone produces digest uniformity at K. The carry channel produces an additional layer of suppression specifically at K. Neither layer alone is required, but the carry-exhaust differential records the K position as a structural locus even when the digest does not.

6. The Aperture Firewall Theorem

We can now state the unified theorem in its full form.

6.1 Statement

Theorem (Aperture Firewall). *Let S be the SHA-256 schedule, viewed as a meet-semilattice over lineage sets. Then:*

(i) S has a unique balanced bottom K = {16, 17} with the property that removing or identifying K collapses the dual-rail structure (D = 0) and is the only single pair-intervention that produces the balanced collapse signature (0, 2, 0, 0, 11).

(ii) After conditioning on schedule position, K-damage and lineage class produce no digest-separable residual signal in Track A or Track B. The position baseline E[HD | t] absorbs the entire raw effect; the residualized class effect is zero (E43) and K is not a residual outlier in either track (E45: |Z| < 0.6).

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(iii) The invisibility of (ii) is not a property of the carry arithmetic alone. It holds independently in the GF(2) carry-free shadow. However, the differential channel Δ = Track A − Track B retains a K-specific carry-exhaust signature ($Z(t = 16) = -3.33$). The carry channel records the seam even while the digest boundary erases it.

In plain language:

The SHA-256 message schedule contains a dual-rail internal topology whose minimum sampling aperture is K = {16, 17}. Damaging or identifying K uniquely collapses the rail structure, but this collapse is not visible as a digest-level Hamming-distance anomaly after position normalization. The invisibility holds both in real SHA-256 and in the GF(2) carry-free shadow. The carry-exhaust differential Δ = A − B nevertheless detects K, showing that the carry channel records the seam even while the digest boundary erases it.

6.2 The Triangle

The theorem corresponds to a single geometric figure. The dual wave has three terms: BUILT (W[16], the even-rail root), UNBUILT (W[17], the odd-rail root), and GAP (the non-joining condition $R8 \vee R9 \notin$ lattice). Closure exists if and only if all three vertices are non-degenerate.

Removing W[16] collapses BUILT. Removing W[17] collapses UNBUILT. Identifying W[17] := W[16] collapses GAP — the two rails become identical, the third vertex disappears, the triangle becomes a line.

The minimum aperture for the dual wave is precisely the triangle's two BUILT vertices, with the GAP induced by the lattice geometry. K = {16, 17} is the unique pair of vertices supporting this triangle, and the engine sweep shows no other pair (i, j) in the seam region produces the same balanced collapse fingerprint.

7. Machine-Speed Trust

The phenomenon characterized here is not unique to SHA-256. Any system whose internal state evolves faster than the observer's readout will produce the same dynamics: load-bearing structure operating below the resolution of direct inspection, externalized only through monitored boundaries. We propose aperture firewall as the name for this class of systems and aperture trust as the name for the corresponding governance posture.

7.1 NIST AI RMF as Aperture Governance

NIST's AI RMF defines trustworthy AI by a set of system characteristics — valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair with harmful bias managed — and organizes risk work through the continuous functions GOVERN, MAP, MEASURE, and MANAGE. The framework already treats accountability and transparency as boundary properties rather than microstate-visibility demands. The Generative AI Profile makes the operational logic more explicit: TEVV applied iteratively, real-time monitoring of performance and trustworthiness characteristics together with alerts for human intervention when deviations arise.

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This is the governance analogue of an aperture firewall. Trust is attached to monitored boundaries, measured invariants, and escalation triggers rather than to the fantasy that a human can watch every internal transition of a machine-speed system.

7.2 The Aperture Trust Posture

Aperture trust is not blind trust, and it is not total visibility. It is calibrated confidence that a system's internal load-bearing structures are being continuously checked at the correct interfaces, with escalation when monitored behavior exits the validated operating aperture. In the SHA-256 case, the schedule seam $K = \{16, 17\}$ is internally indispensable but digest-invisible after correct normalization. In the AI case, the analogue is a model or agent whose internal state transitions cannot be manually inspected in real time but whose operation can still be governed by verified controls, continuous measurement, provenance, and intervention thresholds.

The single-line summary the trust line offers:

As internal structure accelerates beyond direct readout, trust cannot be assigned to the output alone. The trust object becomes the aperture: the smallest internal structure whose removal destroys closure while remaining invisible at the boundary.

8. Open Problems

Three open problems are isolated by this work.

8.1 The Seam Cone

The 11 critical schedule positions $\{0, 2, 9, 10, 14, 15, 16, 17, 18, 19, 20\}$ form a structured set: seam feeders, the seam itself, the rail atoms, and the first downstream extension. We have not given this set an independent algebraic description (e.g. orbit, weight class, residue class mod 210). Such a description would tighten the aperture-firewall theorem from a structural observation to a derivable identity. Conjecture: the seam cone admits a wheel-210 reading consistent with the residue alphabet established in E37.

8.2 The t = 20 / t = 21 Boundary

Position 20 is critical (D drops to 1 on removal); position 21 is safe (D preserved). The lag-7 arm reaches $W[16]$ at $t = 23$ — $W[23] = \sigma_1(W[21]) + W[16] + \ldots$, so by $t = 21$ the rail structure becomes redundantly accessible. Conjecture: position 21 marks the completion of dual-wave bootstrap, after which the rails are protected by redundant inheritance paths. A formal proof would identify the minimum redundancy depth as a function of lag offsets.

8.3 Generalization to SHA-512 and Keccak

SHA-512's schedule has the same recurrence shape with different lag offsets ($W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16]$, over 64-bit words and 80 rounds). The dual-wave / minimum-aperture question can be asked directly. For Keccak, the analogous question is whether the squeeze boundary contains a comparable aperture pair. This would extend the firewall theorem from one standard to a family.

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9. Conclusion

We have shown that the SHA-256 message schedule contains a structured internal object — a graded meet-semilattice on 11 residue classes with a uniquely balanced bottom K = {16, 17} and two non-joining parity horns — and that this internal organization is invisible at the digest boundary after the obvious position confound is removed. K = {16, 17} is the unique minimum sampling aperture for the dual wave: only that pair-removal produces the balanced collapse signature, and no downstream perturbation reproduces it. The firewall is scaffold-stable and carry-reinforced: the invisibility holds in both real SHA-256 and the carry-free GF(2) shadow, while the carry-exhaust differential A − B alone retains the K signature. The result is conservative: it does not claim SHA-256 weakness, digest leakage, or backdoor structure. It claims, and demonstrates, that a hash function can be internally structured, externally uniform, and structurally trustworthy in a precise sense — the aperture firewall.

The broader implication is governance-shaped. Faster systems do not eliminate trust; they force trust to become more formal. In cryptography that means respecting the difference between internal load-bearing geometry and output observability. In AI governance it means building trust from verified invariants, monitored boundaries, and bounded intervention — exactly the structural form the NIST RMF already gestures toward. The aperture is the trust object.

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